Terminal arithmetic in coefficient construction

T-1892 · Prime-field coefficient path · 2026-11-08

The scalar power helper skips its final unused square. A shared reverse-prefix helper avoids multiplication by the leading prefix1 and the final unused accumulator update. Compiled coefficients, ordinary/shared streams and progression evaluation use the new ending.

Remove work after the last needed value

Power: consume bit ⟶ shift exponent

Square only if another bit remains

Prefix inverse: process indices N−1 … 1

Write accumulated inverse directly at index 0

Retain caller inverse checks; test empty/singleton batches and exponent zero.

Power identity. Positive exponent n now uses bit_length(n)−1+popcount(n) multiplication calls. Exponent0 retains zero calls. An independent fixed64-bit left-to-right oracle checks values, including full-width exponents.

Reciprocal identity. Before the final index, the accumulator already equals the first denominator's inverse. Assign it directly. The tail needs 2·max(N−1,0) multiplications; the caller's prefix construction and inverse verification remain. The compiled path also reuses its denominator vector.

Qualification. Focused tests exercise 17,556 powers, 96 batches and 464 inverse values, with three malformed-prefix controls. Shared progression qualification remains: 234 setups and 151,020 evaluation values. The native receipt records inherited recovery output equality separately.

Accounting. A positive power saves one multiplication; a nonempty integrated reciprocal tail saves two. A batch using both changes saves three by source derivation. Fused/neighbor-specific helpers and source-field arithmetic are separate; no full-recovery or runtime saving is inferred from this local count.

Evidence. Reversible source edits and compiler dependency hashes bind the implementation. The exponent95 instrument control is explicitly amended from13 to12 products; old evidence is preserved. Producer assertions await independent signed review. No comparative panel, cloud or timed workload ran.

Next step. Rebind the matched width panel to a common updated implementation; retain source setup and outside-engine arithmetic as separate costs. Further structural compression remains part of the active research objective.